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Thomas Keane, MD, MBA
Assistant Secretary for Technology Policy
Department of Health and Human Services
ASTP/ONC
Attn.: HHS Health Sector AI RFI
Mary E. Switzer Building, Mail Stop: 7033A
330C Street, SW
Washington, DC 20201

RIN 0955-AA13: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dear Dr. Keane:

On behalf of its membership, the Pharmacy Health Information Technology Collaborative (PHIT) is pleased to submit comments for the *Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care***.**

PHIT has been involved with the federal agencies, including the Department of Health and Human Services (HHS) Assistant Secretary for Technology Policy/Office of the National Coordinator (ASTP/ONC) and the Centers for Medicare & Medicaid Services (CMS), in developing the national health information technology (HIT) framework for implementing secure access of electronic health information to improve health outcomes since 2010.

Pharmacists provide essential, patient-centered care services to their patients, including Medicare and Medicaid beneficiaries. Pharmacists use health IT, provider directories, telehealth, e-prescribing (eRx), electronic medical record (EMR)/electronic health record (EHR) systems, and certified EHR technology (CEHRT) to help manage patients' health needs. PHIT supports the use of these systems, which are important to pharmacists in working with other health care providers to provide longitudinal person-centered care planning, needed medications, and transmit patient information related to overall patient care, transitions of care, immunization, medication lists, medication allergies, allergy reactions, patient problem lists, smoking status, and social determinants of health (SDOH). Pharmacists also use health IT for reporting to public health agencies (e.g., immunization reporting), clinical decision support services/knowledge artifacts, drug formulary checking, comprehensive medication management (CMM), and are starting to use artificial intelligence (AI) to enhance efficiency, safety, and person-centered clinical care.

Our comments pertain primarily to question #1, regarding barriers to AI's use in clinical care, the issues and challenges, and presenting possible solutions.

Comments

PHIT supports the development and use of artificial intelligence (AI) that maintains patient safety, privacy, equity, and public trust for use in pharmacy clinical care settings. Additionally, PHIT supports a risk-proportionate framework that tailors appropriate regulatory oversight with compliance requirements, while recognizing that regulations may not address all problems that arise or potential ones. As AI for clinical care evolves, other solutions to potential downstream issues may present themselves and not necessarily require regulatory intervention.

Because AI and digital innovations provide an incredible opportunity for clinical care, PHIT established a Clinical Workflow, Artificial Intelligence, and Digital Innovation (CAD) Workgroup to improve quality patient care across the continuum by leveraging digital innovation and AI at the intersection of clinical workflows and electronic standards (e.g., algorithmic clinical decision support (CDS), machine learning, augmented AI-human intelligence, etc.). The workgroup assists PHIT in developing guidance for pharmacists and key stakeholders about digital health, clinical aspects of e-prescribing, and workflow related to the pharmacists' patient care process (e.g., "Artificial Intelligence Considerations for Pharmacists").

Some AI models are in use currently in various pharmacy settings for improving medication safety and dispensing (e.g., identifying medication orders and prescriptions with common errors, flagging interactions, and enabling the pharmacist to intervene before the medication is dispensed), treatment protocols, enhancing clinical decision through clinical decision support systems (CDSS), patient adherence, and operational efficiency. One example is the National Council for Prescription Drug Program's (NCPDP) Real-Time Prescription Benefit (RTPB) standard that allows real-time transactions of patient-specific drug coverage, cost, and formulary alternatives at the point of prescribing. This minimizes delays, improves medication adherence, and provides accurate out-of-pocket estimates. It is a national AI ready infrastructure that could be leveraged to help clinical care and advance HHS's AI strategy (see also "Real Time Benefits Check: Improving the patient experience and reducing costs.")

Additionally, electronic prior authorization (ePA) in pharmacy is also rapidly evolving from a simple digital form-submission tool into an AI-powered, automated process. These systems are designed to eliminate the administrative burden of traditional fax/phone, reducing prior authorization approval times from days to minutes by using AI to pre-populate forms, analyze clinical data, and predict, rather than just submit, approvals (see also "Enhancing Prior Authorization with Automation and AI-Powered Insights").

Simultaneously, models with applications in clinical pharmacy practice for use in the near future (e.g., disease treatment outcome prediction, clinical documentation and patient chart analysis, pharmacovigilence, and patient education) are being developed, studied, and tested.

Although the adoption of AI in pharmacy practice brings opportunities, it also brings challenges, risks, and barriers. Accelerating the adoption and use of AI can offer significant benefits, provided regulations and guardrails needed to ensure the integrity and safety of data and patients remain in place and are not removed. There is some concern about the proposed HTI-5 (Health Data, Technology, and Interoperability) rule with regard to reducing and removing some AI certification criteria. As AI is a tool that needs appropriate and ethical management, AI adoption must be done in a responsible way.

1. What are the biggest barriers to private sector innovation in AI for health care and its use in clinical care?

The larger barriers affecting AI use in clinical care include:

- **Lack of proper infrastructure**. Many pharmacies, as well as other health care providers, do not have the specialized hardware needed for training and running AI efficiently. Basic AI tasks can run on standard CPUs (central processing units); however, heavy workloads for deep learning require expensive, high-performance hardware with specialized GPUs (graphic processing units). Incentivizing pharmacists through PIPs could help in securing needed specialized hardware (see Interoperability comment below).

- **Interoperability.** Not all existing pharmacy clinical care systems can connect with AI technologies. Pharmacy systems have been moving toward interoperability, though it is not fully where it needs to be. One reason for this is pharmacies were not included in, nor incentivized, as numerous other health care providers were, through the Centers for Medicare & Medicaid Services (CMS) Promoting Interoperability Programs (PIPs), formerly the EHR Incentive Programs or "Meaningful Use."

  "Although electronic health record (EHR) was considered from the beginning of its use, pharmacy software systems have remained largely on the sidelines of these conversations, and as a result, their current state of interoperability has been limited."[1] A 2023 pharmacy data interoperability report by Leavitt Partners "found that many pharmacy stakeholders have inconsistent capacities for importing clinical data to create patient profiles and analyze clinical documentation. While some pharmacies have found innovative solutions for viewing clinical data from other providers, other pharmacies have had to use outdated modalities such as email, fax, or mail. Because of these technical limitations in interoperability, pharmacists are hindered in their ability to deliver clinical care they were trained to provide, further frustrating data interoperability modernization."[2] Interoperability limitations also lead to fragmented data (see also Data quality comment below).

  The Leavitt report also recommends that CMS include "pharmacies in existing and emerging value-based care models" and "incentivize pharmacies to invest further in data

[1]Kenneth C. Hohmeier, Jason Ausli, Laila Hajyani, et al, "Early implementation of Pharmacist eCare Plan and electronic health record integration: A proof-of-concept study on health information interoperability," *JAPhA Practice Innovations*, October 2025.
[2] "Supporting Pharmacy Data Interoperability: An Imperative for Patient Access and Outcomes," Leavitt Partners, April 2023, page 6.

interoperability by enabling them to contribute to patient value and share in revenue as members of the interdisciplinary care team. This step could encourage pharmacy data interoperability within existing statutory definitions of 'provider.'"[3]

Including pharmacists in PIPs' incentives would help strengthen interoperability and ensure that pharmacists can obtain relevant clinical information, laboratory data, medication changes, diagnoses, etc., in real time.

- **Workforce challenges and skills gaps**. AI adoption appears to be outpacing workforce capability. Recent reports indicate that although AI adoption and usage in health care increased in 2025, they highlight a significant gap between adoption rates and employer training. A 2024 survey showed only 24% of health care workers received employer-provided AI training, even though they are using AI tools.[4] This survey also revealed that "AI in health care is primarily used for administrative tasks, while more clinical applications are still in the early stages of adoption."[5]

- **Contract clarity – liability.** With the administration currently moving forward with loosening and removing regulations to accelerate the development and adoption of AI, liability allocation within contracts with AI health solution vendors becomes an even greater concern that could impact the adoption of AI in health care settings, particularly pharmacies. AI-generated content and recommendations influence medical and health care decisions. Continued concern pertains to who is responsible if something goes wrong regarding an AI recommendation.

  "AI vendors position themselves as only a tool to health care providers and try to disclaim liability for any downstream use."[6] Health care providers expect vendors to stand behind their products, especially those products that generate clinical notes, diagnostic suggestions, etc.[7]

  The concern is that if regulations are loosened too much or removed, AI developers and vendors will further limit their liability for their products and the content generated, making health care providers the more liable because they are the users. Additionally, insurers are starting to pivot away from AI liability. Three major insurers – AIG, Great American, and WR Berkley – "recently submitted requests for regulatory approval to limit their liability for claims arising from artificial intelligence systems, including chatbots and other automated services."[8]

  If regulations are overly loosened or removed, pharmacists will need to try and negotiate with AI vendors for specific indemnification for risks, such as algorithmic bias and discrimination, data privacy breaches, incorrect AI-generated outputs (e.g., AI-specific

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[3] Ibid., page 12.
[4] "Medscape and HIMSS Release 2024 Report on AI Adoption in Healthcare," PRNewsire, December 6, 2024.
[5] Ibid.
[6] "AI Contracts in Health Care: Avoiding the Data Dumpster Fire," Foley, June 26, 2025.
[7] Ibid.
[8] "Major Insurers are Pulling Back from AI Liability," Metropolitan Risk blog, November 24, 2025.

errors), data use and training rights, and data leakage. AI developers and vendors tend not to be open to assuming more liability; they prefer less liability.

If the goal is to accelerate the adoption of AI in health care, ASTP then needs to help health care providers by developing model language for use in contracts so that shared liability is equitable, as well as issuing risk-tier rules regarding liability allocation.

- **Data quality – siloed and fragmented.** Health data is often siloed and fragmented, many times because of interoperability limitations, making it not only hard to train AI models but also preventing health care decision-makers (including pharmacists) from seeing the full picture. Siloed data, known as the silent killer of AI initiatives, reduces AI model accuracy and reliability. Generative and predictive AI models trained on partial data cannot understand the full context of a problem, which could lead to poor and inaccurate health recommendations from AI. Data and recommendations from AI in clinical care need to be trusted.

  Many pharmacy patients see multiple providers that prescribe medications and use various payment systems (e.g., pharmacy benefit managers (PBMs), health plans) that create siloed and fragmented data.

  "Because no standardized or universally available data-sharing solution exists for pharmacies, health plans have implemented a number of workarounds. While pharmacy systems seamlessly share data related to prescription drug claims between pharmacies and health plans, the same infrastructure has not been applied for sharing clinical services or clinical patient data. Therefore, when health plans and pharmacies collaborate on clinical initiatives, such as transitions of care programs, chronic disease management or other initiatives, data-sharing workarounds must be relied upon."[9]

  The result is a pharmacist working with "multiple plans on multiple patient care initiatives may have to document via various fragmented and disconnected systems that require varying types and formats of information to be entered, often manually."[10]

- **Data and algorithmic bias.** AI algorithms used to treat and diagnose patients can have biases. Such bias in AI arises when it is trained on biased data. Although it may not be intentional, medical data contain bias, which could perpetuate historical bias.[11] Bias that occurs in health care settings is generally found in patient notes. If data from those notes are given to an AI model, the model will learn and reproduce those biases.[12]

  Data bias increases the risk of discrimination in health care. If data lack diversity (e.g., age, gender, ethnicity, geographic locations, socioeconomic status, language, customs, cultural, etc.) and is used for AI training, then the data are misrepresentative of the full

[9] Ibid., page 9.
[10] Ibid., page 9.
[11] Sian Carey, Allan Pang, Marc de Kamps, "Fairness in AI for healthcare," *Future Healthcare Journal*, Volume 11, Issue 3, September 2024.
[12] Ibid.

population.[13] Most health care data used to train AI models for use in the United States comes from patients in the U.S., China, and parts of Europe.[14] Not all patient data used to train AI is representative of the U.S. population. "AI systems trained on biased data often perform poorly when applied to populations with different genetic backgrounds, disease prevalence, or environmental exposures."[15]

Health care and health care systems will face challenges in addressing algorithmic biases, especially as it relates to medications. AI for clinical care should be held to standardized outcomes-focused metrics to ensure accuracy and to reduce risks that may cause patient harm.

- **Data privacy and security concerns**. AI relies on vast amounts of patient data. Collecting vast amounts of data creates privacy and security challenges. Health care practice settings, including pharmacies, using AI are more vulnerable to data breaches and cyber-attacks targeting patient information, drug data, etc., particularly as more pharmacists and other health care providers use chatbots, such as ChatGPT, in health care.[16] "Using ChatGPT can put users at risk of losing control of sensitive patient information, source code, and more,"[17] especially if patient identifiable information is inadvertently provided. As chatbots have become the go-to interface for interacting with large language models (LLMs), not one LLM does it all. This means multiple chatbots may likely be used, further increasing data privacy and security risks.

  Those adopting AI tools in health care will need to ensure the tools they use comply with the Health Insurance Portability and Accountability Act (HIPAA), as the data being used by AI may likely include protected health information (PHI), personal data, and health data. In this regard, AI developers and vendors need oversight.

- **Expertise replacement.** Relying solely on AI, without incorporating augmented AI, for health care recommendations presents safety, clinical care, and liability risks. Pharmacists play a vital role in bridging the gap between technology and personalized care. Integrating AI into clinical care should not overshadow the human element of care; augmented AI as the assistive role of AI can help in this regard. AI systems can make mistakes, and without human oversight, errors can result in patient harm. Safeguards must be put into place to ensure that a human verifies the accuracy and completeness of AI recommendations that impact patient care.

  Among the AI adoption and usage training needed in the health care workforce are: data literacy (i.e., how to interpret AI-driven data, evaluate algorithms, and recognize potential biases), understanding AI bias in AI models to prevent inequity in patient care, managing human-AI collaboration, auditing AI-generated content (e.g., drug interaction

[13] Kathleen Kenny, PharmD, RPh, "Carefully Consider Ethical Implications of Artificial Intelligence Use in Pharmacy, *Pharmacy Times*, November 29, 2025.
[14] Gabriel Onuh, "AI in healthcare risks could exclude 5 billion people; here's what we know about it," World Economic Forum, October 2025.
[15] Ibid.
[16] Li J, "Security Implications of AI Chatbots in Health Care," *J Med Internet Res* 2023; 25:e47551.
[17] Ibid.

checks, dosage recommendations), and integrating AI into clinical practice, especially into existing workflows.

Training is also needed in data privacy, patient safety, and ethical considerations to avoid over-reliance on technology.

- **Costs.** AI adoption is an expensive investment and may be prohibitive for some health care providers, especially small providers. Although small-scale basic AI systems start average between $10,000-$50,000, total implementation often start at $1 million and can exceed $10 million.[18] Mid-sized is $100,000-$500,000, while enterprise grade AI solutions can be $1 million-$10 million.[19] These costs are for initial implementation, not ongoing operational and maintenance costs (average 15-30% of the original build cost); retraining and model updates (AI is not static); infrastructure usage increases; and security, compliance, and risk management.[20]

- **Trust and ethics.** This is the top-tier concern and barrier in AI adoption. Clinician trust in AI and machine learning (ML)-based predictive DSI (differentiable search indexing) will be crucial for adoption. A lack of trust comes from AI associated risks, such as bias and fairness, data privacy, accountability and security, and limited transparency. As AI becomes integrated into decision-making processes, ethical implications and concerns are arising. Studies indicate that people lack trust in AI to make ethical decisions, which is critical in health care, especially for patient safety.

  Deloitte's most recent survey of 1,800 businesses and technical professionals globally, conducted in 2024, showed that 54% of respondents said AI "posed the most severe ethical risk among emerging technologies, with top concerns for GenAI being data privacy, transparency, and data provenance."[21] Further, the aforementioned study published by *JAPhA* indicated that 34.9% of pharmacists expressed some degree of distrust of AI.[22]

  AI faces many ethical dilemmas, especially in health care, as it is not inherently ethical (lacks moral agency). The use of AI in health care raises ethical concerns related to patient privacy, data, recommendations for treatment provided by diagnostic tools, and the potential for AI to replace human expertise.

  AI can be designed and taught to produce ethical, fair, and transparent outcomes, provided ethical training is built in. To do that, AI developers must proactively mitigate bias in training data; create algorithms that are interpretable, allowing humans to understand the reasoning behind AI decisions; and be human-centric, aligning with human values.

  To build trust in AI, however, government agencies need to be careful in loosening and removing regulations and other guardrails designed to ensure the development and use of

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[18] Lina Taposhi, "The True Cost of Implementing AI in Business in 2026," Rise Up Labs.
[19] Ibid.
[20] Ibid.
[21] "State of Ethics and Trust in Technology Report," *Deloitte*, September 2024.
[22] Ibid., "Pharmacists' perceptions," *JAPhA*.

AI in health care in general, clinical care specifically, is safe and does not harm patients. Trust is crucial because AI systems will be making decisions that can significantly impact individuals and society.

Patient safety and safeguarding public health need to be the top priority in AI-health care development.

2. What HHS regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care?

Pharmacists need to be included in the CMS PIPs incentives as eligible health professionals. When the program was launched in 2011 as the EHR Incentive Programs or "Meaningful Use" to promote the adoption of and meaningful use of Certified Electronic Health Record Technology (CEHRT), pharmacists were excluded from the program. PHIT has requested numerous times since the program's inception that the Secretary remedy this oversight.

Although pharmacists have adopted EHR technology without incentives, as other health care professionals have been afforded, not being included in the incentive program is one of the reasons achieving interoperability in pharmacy settings has been slow and not fully where it needs to be.

With the rapid advancement and integration of AI into pharmacy practice, PHIT strongly believes it is time to include pharmacists in PIPs for needed incentives to adopt AI and so that full interoperability can finally be realized.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

- Develop an AI toolkit to assist health care providers to safely integrate AI into clinical care. ASTP currently provides tool kits for health care providers that support effective implementation of health IT, interoperability, patient engagement, and EHR.
- Reactivate the ASTP Health IT Advisory Committee's (HITAC) Pharmacy Interoperability and Emerging Therapeutics Task Force to help address concerns and make further recommendations to move AI adoption forward. In 2023, the pharmacy task force put forward Recommendation 13 that included an AI component.
- Continue certification for developers of health IT, particularly AI, ML, deep learning AI, etc. Certification would increase trust in AI.
- Because loosening and removing regulations will impact contracts between AI vendors and health care providers, ASTP should develop model language for providers to use so that shared liability is equitable. This could be part of an AI toolkit (noted above). As mentioned previously, with the forthcoming changes from HTI-5, AI vendors will likely seek to further limit their liability, making health care providers more liable because they are users of the AI vendors' products (see above "Contract clarity – liability" comments).

- Establish a harmonized, risk-proportionate federal framework across federal health agencies and state-level requirements. A unified framework would provide clarity and support AI deployment in clinical care settings, including pharmacies.
- Include pharmacists in PIPs as eligible health professionals for those incentives, which will help alleviate interoperability limitations. Pharmacists are consistently ranked among the top three most trusted professionals in the United States, and they provide a range of extensive clinical care services, especially to Medicare patients. In many rural areas, pharmacists are often the only immediate accessible health care providers.
- Invest in public-private partnerships and demonstration projects that validate AI clinical care referral models in real-world pharmacy settings.

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The Pharmacy HIT Collaborative comprises the major national pharmacy associations, representing 250,000 members. PHIT's membership is composed of the key national pharmacy associations involved in health IT, the National Council for Prescription Drug Programs, and associate members encompassing e-prescribing, health information networks, transaction processing networks, pharmacy companies, system vendors, pharmaceutical manufacturers, and other organizations that support pharmacists' services.

As the leading authority in pharmacy health information technology, PHIT's vision and mission are to ensure the U.S. health IT infrastructure better enables pharmacists to optimize person-centered care. Supporting and advancing the use, usability, and interoperability of health IT by pharmacists for person-centered care, PHIT identifies and voices the health IT needs of pharmacists; promotes awareness of functionality and pharmacists' use of health IT; provides resources, guidance, and support for the adoption and implementation of standards-driven health IT; and guides health IT standards development to address pharmacists' needs. For additional information, visit www.pharmacyhit.org.

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On behalf of PHIT, thank you again for the opportunity to comment on the *Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care*

For more information, contact Jeff Rochon, Executive Director, Pharmacy HIT Collaborative, at jeff@pharmacyhit.org.

Respectfully submitted,

Jeff Rochon, Pharm.D.
Executive Director
Pharmacy HIT Collaborative
jeff@pharmacyhit.org

Chad Worz, PharmD, BCGP
Chief Executive Officer
American Society of Consultant Pharmacists (ASCP)
cworz@ascp.com

Ronna B. Hauser, PharmD
Senior VP, Policy & Pharmacy Affairs
National Community Pharmacists Association (NCPA)
ronna.hauser@ncpa.org

Michael D. Hogue, PharmD, FAPhA, FNAP, FFIP
Executive Vice President and CEO
American Pharmacists Association (APhA)
mhogue@aphanet.org

Scott Anderson, PharmD, MS, CPHIMS,
FASHP, FVSHP
Director, Member Relations
American Society of Health-System Pharmacists (ASHP)
SAnderson@ashp.org

Kevin Nicholson, R.Ph., JD,
Vice President of Policy, Regulatory, and Legal Affairs
National Association of Chain Drug Stores(NACDS)
KNicholson@NACDS.org

Kathy Pham, Pharm.D., BCPPS
Senior Director, Policy and Professional Affairs
American College of Clinical Pharmacy (ACCP)
kpham@accp.com